

Joy (Yu Ting) Chen

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Experiences

Qualcomm

Santa Clara, California, USA

Sr Software Engineer

Jan 2023 – Dec 2025

- Worked in the Compiler Toolchain team behind the Snapdragon X Elite CPU.
- Developing ML workloads and supporting hardware team in gathering traces for performance measurements.
- Created visualization tool for viewing different addressing modes of new hardware registers.
- Debugged performance analysis bugs using various profiling tools.

Tenstorrent Inc

Toronto, Ontario, Canada

Sr Software Engineer

Oct 2019 – Dec 2022

- Sped up compilation time by 23x on large transformer networks for custom ML compiler by implementing compilation caching system.
- Created initial implementation of runtime software to integrate software stack with PyTorch frontend.
- Co-led the exploration of training transformer models with dynamic runtime sparsity features. Implemented using PyTorch with three patents pending.
- Set up training flow using AWS and other cloud services. Work included entire process from training data tokenization, managing storage on cloud services through S3, tracking performance of models on W&B, fine-tuning on specific language tasks.
- Created a highly efficient graph library for custom ML compiler using C++ with Python interface similar to networkx graph library using functional programming principles. Library adoption showed a speedup of $\sim 100\times$.

LegUp

Toronto, Ontario, Canada

Open-source contributor

2013 – 2016

- LegUp is an open-source project based in LLVM, it was started in 2011 at the University of Toronto. In 2016, LegUp was incorporated as LegUp Computing, and acquired by Microchip in 2020.
- Implemented memory partitioning to enable higher block RAM access throughput. This work won the best paper award at FPL 2017.
- Implemented control flow merging in order to parallelize compute on FPGA.

Intel Corp

Hudson, Massachusetts, USA

Graduate Software Intern

Feb 2016 – Jun 2016

- Created open-source analysis framework to find code regions amenable to hardware acceleration for heterogeneous computing platforms through analysis of dynamic dataflow graph constructed based on program execution.
- Led initial research and development culminating in final presentation demonstrating proof-of-concept tool to multiple groups within Intel.

Skills

Programming Languages: C++, C, Python, HTML/CSS/Javascript

Major Libraries & Tools: PyTorch, Pybind, LLVM

Domains: Compiler, Machine Learning, Algorithms & Data Structures, FPGA

Education

University of Toronto

PhD in Electrical and Computer Engineering

- Visualization for hardware design CAD tools.

Toronto, Ontario, Canada

Sep 2017 – Not completed

University of Toronto

Masters in Electrical and Computer Engineering

- **Y.-T. Chen**, J.H. Anderson, “Automated Generation of Banked Memory Architectures in the High-Level Synthesis of Multi-Threaded Software”, Int’l Conference on Field-Programmable Logic and Applications (FPL), Ghent, Belgium, Sep 2017. (Best Paper Award)
- R. Nane, V. Sima, C. Pilato, J. Choi, B. Fort, A. Canis, **Y.T. Chen**, H. Hsiao, S. Brown, F. Ferrandi, J. Anderson, K. Bertels, “A Survey and Comparison of FPGA High-Level Synthesis Tools”, IEEE Trans. on Computer-Aided Design of Integrated Circuits and Systems (TCADS), 2015.
- A. Canis, J. Choi, B. Fort, B. Syrowik, R.L. Lian, **Y.-T. Chen**, H. Hsiao, J. Goeders, S. Brown, J.H. Anderson, “LegUp high-level synthesis”, chapter in FPGAs for Software Engineers, Springer, 2015.
- B. Fort, A. Canis, J. Choi, N. Calagar, R. Lian, S. Hadjis, **Y.T. Chen**, M. Hall, B. Syrowik, T. Czajkowski, S.D. Brown, J.H. Anderson, “Automating the Design of Processor/Accelerator Embedded Systems with LegUp High-Level Synthesis”, Int’l Conference on Embedded and Ubiquitous Computing (EUC), Milan, Italy, August 2014. (Invited paper).

Toronto, Ontario, Canada

Sep 2014 – Jul 2017

University of Toronto

Bachelors in Engineering Science, Electrical and Computers

Engineering Major

- 16 month internship at AMD.

Toronto, Ontario, Canada

Sep 2009 – Jun 2014

Awards

2018 - **NSERC CGS Postgraduate Scholarship**, National level scholarship
2022 for PhD students in Canada. Amount C\$70000.

NSERC

2018 **Graduate Scholarship**, Departmental award. Award amount
C\$4350.

Right Track
CAD

2017 **Michal Servit Best Paper Award**, Awarded for best
paper/presentation at International FPGA conference. Award
amount €500.

Int’l
Conference
FPL